At the Laboratory for Emerging Intelligence at UCSD, my overarching goal is to understand and improve Large Language Models' (LLMs) task-solving capabilities **through the lens of information retrieval (IR), extraction (IE), and utilization**. To clarify, my research focused on the fundamental question: **how effectively can LLMs process and utilize new information at inference time to solve tasks?** Specifically, I have investigated the following questions: (1) Can LLMs identify and connect pieces of useful information, especially when these pieces of information have complex interconnecting dependencies and are scattered across long input documents? (2) Can state-space models (SSMs) serve as a more efficient and accurate alternative to transformers for processing and retrieving interdependent information from long documents?

My research studies the interdependencies between multiple pieces of information, which fundamentally differ from traditional IR and IE. For example, consider a question that requires two pieces of information (A and B) to answer. A and B are separated by hundreds of thousands of tokens in the input text. B depends entirely on A, and you cannot understand B or identify its relevance to the question unless you first find A. My research demonstrates that long documents contain even more complex information dependencies beyond what IR/IE methods like embedding models can handle, and capturing these dependencies is crucial for LLMs to effectively answer questions and solve problems based on long input documents.

In the following sections, I will present five completed research papers along two axes: (1) developing models for long document question-answering and IR; (2) creating novel benchmarks for communities working on IR, scientific document understanding, BioNLP, and AI Scientist.

I led a team of five researchers in the Mamba Retriever project to tackle the task of long-document question-answering. While prior works use transformer-based embedding models to identify relevant information from long documents in Retrieval-Augmented Generation (RAG) settings, I noticed a major limitation for embedding models–they focus on isolated text segments without utilizing global context. Inspired by SSMs' linear inference time and recurrent architecture, which enable efficient processing of long documents using fewer FLOPS and maintaining global-contextual awareness, I designed a sentence-level classification head for the Mamba SSM to retrieve question-relevant sentences. My main challenge was teaching this model to capture long-range dependencies between sentences using global context. My key insight was to generate synthetic training data to simulate long-range dependencies between sentences. However, I encountered another challenge: any synthetically generated dependencies between sentences were unnatural and ineffective. I finally realized that I must use real dependencies between sentences as training data. I then designed a multi-stage pipeline that leverages LLMs to discover real semantic connections between distant sentences. The resulting Mamba retriever outperformed state-of-the-art embedding models across 41 long-document QA benchmarks while being faster and using fewer FLOPS. My work demonstrates how improving training data quality can be more impactful than architectural changes. Our paper, "Efficient Full-Context Retrieval for Long Documents", received positive ratings (6/6/6/5) upon submission to ICLR 2025.

To train embedding-based IR models, I designed an LLM-powered pipeline that reduced the amount of accurate information used to generate synthetic queries. This approach intentionally

triggered LLM hallucination, which introduced flaws into synthetic queries, thereby making the relationship between queries and documents less obvious. These imperfect pairs proved more effective for contrastive training, as they forced embedding models to focus on deeper semantic relationships rather than overfitting on superficial textual similarities. I learned that synthetic training data need not mirror the testing distribution; instead, it can be strategically designed to cultivate specific model learning behaviors and skills. Our paper "IR2: Information Regularization for Information Retrieval" is published at LREC-COLING 2024 as an oral paper.

I also explored the potential of using LLMs' reasoning abilities for direct information retrieval. As first-author, I developed the BIRCO benchmark to evaluate IR systems on complex, objective-oriented and instruction-following tasks, and showed that LLM-based IR systems are significantly better at following user instructions than embedding models, though incurring higher computational costs. BIRCO is being integrated into MMTEB.

As first-author, I built two novel biomedical benchmarks, EvidenceBench and RoBBR, evaluating models' abilities to extract evidence from biomedical papers - the former for hypothesis support and the latter for risk-of-bias judgments. Benchmarking shows current LLMs fail to extract sentences with complex interdependencies across long documents. Traditionally, creating such benchmarks requires extensive annotation from thousands of domain experts. However, I developed an innovative solution by leveraging existing resources: review papers containing human expert-written summaries of biomedical evidence, and Cochrane meta-analyses documenting human expert-written support judgments for risk-of-bias determinations. I designed LLM-powered pipelines that could decompose these pre-existing expert "annotations" and trace them back to original sentences in biomedical papers, effectively reverse-engineering human experts' decision-making processes. My technique of harnessing existing human "annotations" to create benchmarks using LLM pipelines serves as a valuable precedent for modern benchmark construction efforts. "EvidenceBench: A Benchmark for Extracting Evidence from Biomedical Papers" and "Measuring Risk of Bias in Biomedical Reports: The RoBBR Benchmark" received positive evaluations at NeurIPS 2024, scoring (6/6/6) and (7/6/6) respectively.

I am currently pursuing two research projects. First, building upon RoBBR and EvidenceBench, I am developing a large-scale research knowledge graph that connects scientific hypotheses with each other and their supporting evidence to power AI-Scientist applications. Second, I am investigating the theoretical mechanisms by which LLMs identify and utilize interdependent information across long contexts, with the goal of designing more optimized systems.